

Optimization-Based Beam Selection for SINR, Spectral Efficiency and Energy Efficiency in 6G Massive MIMO Systems

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Abstract—The increasing number of users and antenna elements in 6G wireless networks requires efficient beam management techniques to achieve high spectral and energy efficiency. Hybrid beamforming reduces the number of RF chains compared with fully digital architectures, but assigning a limited number of RF chains to appropriate user-beam combinations remains a challenging selection problem. This work investigates a beam-to-RF-chain assignment framework for a multi-user massive MIMO system. A geometric wireless channel model is first constructed to characterize the propagation between the base station and users. Based on the resulting channel characteristics, a user-beam gain matrix is generated, where each element represents the channel-beam alignment for a particular user and candidate beam. A Greedy beam-selection algorithm is then implemented to iteratively select the highest-gain feasible user-beam pair while ensuring that each selected user and beam is assigned only once. The selected beam assignments are evaluated using signal-to-interference-plus-noise ratio (SINR), spectral efficiency, and energy efficiency. Preliminary simulations are performed for a system with 20 users, 16 transmit antennas, 8 RF chains, and 3 hotspots, with results averaged over 15 independent channel seeds. The obtained results establish a classical Greedy baseline for subsequent beam-selection optimization methods.

Index Terms—6G, Massive MIMO, Hybrid Beamforming, Beam Selection, RF Chains, SINR, Spectral Efficiency, Energy Efficiency, Greedy Optimization

I. CONTRIBUTIONS

The main contributions of this work are as follows:

- **Multi-user massive MIMO channel model:** A multi-user channel model is developed to represent the propagation environment between a multi-antenna base station and users distributed across multiple spatial hotspots.
- **User-beam gain formulation:** A gain matrix is constructed to quantify the suitability of candidate beams for different users based on the alignment between each user's channel and each candidate beamforming vector.
- **Greedy beam-to-RF-chain assignment:** A Greedy selection algorithm is implemented to iteratively select high-gain user-beam pairs while satisfying the one-user/one-beam assignment constraint imposed by the available RF chains.
- **Performance evaluation:** The selected beam assignments are evaluated using SINR, spectral efficiency,

and energy efficiency, establishing a classical baseline for subsequent OMP- and QAOA-based beam-selection methods.

II. PROPOSED WORK

A. System Model

We consider a downlink multi-user mmWave massive MIMO system with $N_t = 16$ transmit antennas serving $K = 20$ users distributed across 3 spatial hotspots. The base station is equipped with $N_{RF} = 8$ RF chains. Since only eight RF chains are available, eight users can be simultaneously assigned distinct beams in the considered beam-selection stage.

Hybrid beamforming is motivated by the hardware limitations of fully digital massive MIMO architectures, where the number of required RF chains grows with the number of antennas [1], [3], [4].

B. Channel Model

The channel between the base station and user k is modeled using a clustered geometric mmWave channel representation [1], [2]:

$$\mathbf{h}_k = \sqrt{\frac{N_t}{N_{cl}N_{ray}}} \sum_{c=1}^{N_{cl}} \sum_{\ell=1}^{N_{ray}} \alpha_{k,c,\ell} \mathbf{a}_t(\phi_{k,c,\ell}), \quad (1)$$

where N_{cl} denotes the number of scattering clusters, N_{ray} denotes the number of propagation rays per cluster, $\alpha_{k,c,\ell}$ is the complex path gain, and $\phi_{k,c,\ell}$ is the angle of departure of the corresponding path. Each user is represented using a single effective receive channel in the present implementation.

The resulting multi-user channel matrix is represented as

$$\mathbf{H} = \begin{bmatrix} \mathbf{h}_1^H \\ \mathbf{h}_2^H \\ \vdots \\ \mathbf{h}_K^H \end{bmatrix} \in \mathbb{C}^{K \times N_t}. \quad (2)$$

For the considered configuration,

$$\mathbf{H} \in \mathbb{C}^{20 \times 16}.$$

C. Transmit Array and Beam Codebook

The base station uses a Uniform Linear Array (ULA) with $N_t = 16$ antennas. For half-wavelength antenna spacing $d = \lambda/2$, the transmit steering vector corresponding to angle ϕ is

$$\mathbf{a}_t(\phi) = \frac{1}{\sqrt{N_t}} [1 \quad e^{j\pi \sin(\phi)} \quad e^{j2\pi \sin(\phi)} \quad \dots \quad e^{j(N_t-1)\pi \sin(\phi)}]^T \quad (3)$$

A DFT-based candidate beam codebook is constructed from the steering vectors [5], [6]. With an oversampling factor of one, the number of candidate beams is

$$M = N_t = 16. \quad (4)$$

The m th candidate beam is therefore given by

$$\mathbf{f}_m = \mathbf{a}_t(\theta_m), \quad m = 1, 2, \dots, M. \quad (5)$$

Thus, the beam-selection problem consists of selecting suitable beams from the $M = 16$ candidate beams for the $K = 20$ users.

D. User-Beam Gain Matrix

The compatibility between user k and candidate beam m is measured using the effective channel-beam alignment:

$$G_{k,m} = |\mathbf{h}_k^H \mathbf{f}_m|. \quad (6)$$

A larger value of $G_{k,m}$ indicates stronger alignment between the channel of user k and candidate beam m . The resulting user-beam gain matrix is

$$\mathbf{G} \in \mathbb{R}^{K \times M}, \quad K = 20, M = 16. \quad (7)$$

E. User-Beam Assignment Formulation

Because only $N_{RF} = 8$ RF chains are available, exactly eight feasible user-beam pairs are selected in the present implementation. Let

$$x_{k,m} \in \{0, 1\}$$

be a binary assignment variable, where $x_{k,m} = 1$ indicates that beam m is assigned to user k .

The gain-maximization formulation is

$$\max_{\mathbf{x}} \sum_{k=1}^K \sum_{m=1}^M G_{k,m} x_{k,m}. \quad (8)$$

The assignment is subject to the following constraints. Each user can receive at most one selected beam:

$$\sum_{m=1}^M x_{k,m} \leq 1, \quad \forall k. \quad (9)$$

Each beam can be assigned to at most one user:

$$\sum_{k=1}^K x_{k,m} \leq 1, \quad \forall m. \quad (10)$$

Exactly N_{RF} user-beam pairs are selected:

$$\sum_{k=1}^K \sum_{m=1}^M x_{k,m} = N_{RF}. \quad (11)$$

The formulation represents the beam-to-user assignment stage of the proposed framework. The present work solves this assignment using a Greedy heuristic, while OMP and QAOA-based optimization are planned for subsequent stages.

F. Simulation Parameters

The main simulation parameters are

$$N_t = 16, \quad K = 20, \quad N_{RF} = 8, \quad M = 16. \quad (12)$$

The users are distributed across 3 spatial hotspots, and the results are averaged over 15 independent channel seeds. Therefore, 20 candidate users and 16 candidate beams are initially available, while 8 user-beam pairs are finally selected according to the available RF chains.

G. Proposed System Architecture

Fig. 1 presents the overall target framework of the project. The currently implemented portion consists of channel generation, beam-codebook construction, user-beam gain calculation, Greedy beam selection, and performance evaluation using SINR, spectral efficiency, and energy efficiency. The subsequent stages will investigate OMP and QAOA-based optimization for the same beam-selection problem.

III. ALGORITHM

Algorithm 1 describes the Greedy beam-to-RF-chain assignment procedure implemented in the present phase.

At each iteration, the algorithm searches the currently feasible user-beam combinations and selects the pair having the highest gain. The selected user and beam are then removed from further consideration. This guarantees that a user is assigned at most one beam and a beam is assigned at most one user. The process is repeated until all N_{RF} RF-chain assignments are obtained.

The computational cost of the direct implementation is

$$O(N_{RF}KM), \quad (13)$$

because the remaining user-beam combinations are scanned at each of the N_{RF} iterations.

IV. IMPLEMENTATION, RESULTS AND PERFORMANCE ANALYSIS

The framework was implemented in Python. The channel model was used to generate multi-user mmWave channels for $K = 20$ users distributed across 3 hotspots, with $N_t = 16$ transmit antennas and $M = 16$ DFT-based candidate beams. The Greedy algorithm was then executed to select $N_{RF} = 8$ feasible user-beam pairs. The reported results were averaged over 15 independent channel seeds.

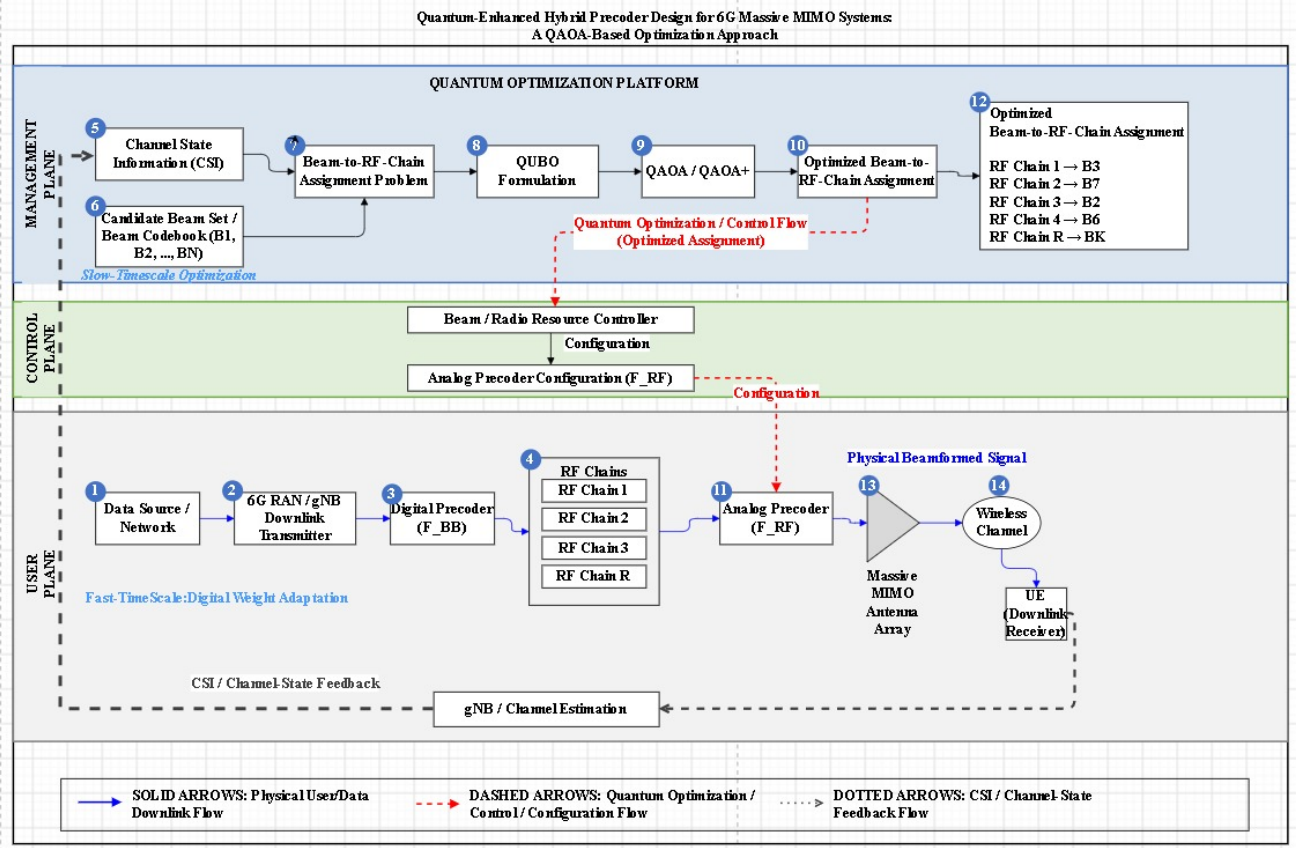


Fig. 1. Overall project framework. The currently implemented channel-model and Greedy beam-selection stages are followed by SINR, spectral-efficiency, and energy-efficiency evaluation. OMP and QAOA-based optimization represent subsequent stages of the project.

A. SINR Evaluation

For the selected user set $\mathcal{S}$, let m_k denote the beam assigned to user k . The received signal-to-interference-plus-noise ratio of user k is expressed as

$$\text{SINR}_k = \frac{P_k |\mathbf{h}_k^H \mathbf{f}_{m_k}|^2}{\sum_{j \in \mathcal{S}, j \neq k} P_j |\mathbf{h}_k^H \mathbf{f}_{m_j}|^2 + \sigma^2}, \quad (14)$$

where P_k is the transmit power allocated to user k and σ^2 is the noise power. The average SINR over the selected users is

$$\overline{\text{SINR}} = \frac{1}{|\mathcal{S}|} \sum_{k \in \mathcal{S}} \text{SINR}_k. \quad (15)$$

The Greedy algorithm selects beams using channel-beam alignment gain rather than directly optimizing the interference term in (14). Therefore, a high individual beam gain does not necessarily result in a high SINR when simultaneously selected beams produce interference at other users.

B. Spectral Efficiency Evaluation

The spectral efficiency of user k is calculated from its SINR as

$$R_k = \log_2 (1 + \text{SINR}_k) \quad \text{bits/s/Hz}. \quad (16)$$

The total spectral efficiency of the simultaneously served users is

$$R_{\text{sum}} = \sum_{k \in \mathcal{S}} \log_2 (1 + \text{SINR}_k). \quad (17)$$

Thus, spectral efficiency measures the total achievable data rate per unit bandwidth for the selected users.

C. Energy Efficiency Evaluation

Energy efficiency measures the achievable spectral efficiency per unit of total consumed power. It is defined as

$$\text{EE} = \frac{R_{\text{sum}}}{P_{\text{total}}} \quad \text{bits/s/Hz/W}, \quad (18)$$

where R_{sum} is the total spectral efficiency and P_{total} represents the total power consumed by the transmission system. The total power can be represented generally as

$$P_{\text{total}} = P_{\text{tx}} + P_{\text{RF}} + P_{\text{PS}} + P_{\text{BB}}, \quad (19)$$

where P_{tx} denotes transmit power, P_{RF} denotes RF-chain power consumption, P_{PS} denotes phase-shifter power consumption, and P_{BB} denotes baseband processing power.

Algorithm 1 Greedy Beam-to-RF-Chain Assignment

Require: Gain matrix $\mathbf{G} \in \mathbb{R}^{K \times M}$; number of RF chains N_{RF}

Ensure: Assignment list of (k, m) pairs; total selected gain

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1: remainingUsers  $\leftarrow \{0, 1, \dots, K-1\}$ 
2: remainingBeams  $\leftarrow \{0, 1, \dots, M-1\}$ 
3: assignment  $\leftarrow \emptyset$ 
4: totalGain  $\leftarrow 0$ 
5: for  $t = 1$  to  $N_{RF}$  do
6:   bestGain  $\leftarrow -\infty$ 
7:   bestPair  $\leftarrow \text{None}$ 
8:   for each  $k \in \text{remainingUsers}$  do
9:     for each  $m \in \text{remainingBeams}$  do
10:      if  $G[k][m] > \text{bestGain}$  then
11:        bestGain  $\leftarrow G[k][m]$ 
12:        bestPair  $\leftarrow (k, m)$ 
13:      end if
14:    end for
15:  end for
16:   $(k^*, m^*) \leftarrow \text{bestPair}$ 
17:  append  $(k^*, m^*)$  to assignment
18:  totalGain  $\leftarrow \text{totalGain} + \text{bestGain}$ 
19:  remove  $k^*$  from remainingUsers
20:  remove  $m^*$  from remainingBeams
21: end for
22: return assignment, totalGain

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TABLE I
GREEDY-ONLY BASELINE PERFORMANCE

Metric	Value	Unit
Average SINR	0.3089	dB
Spectral Efficiency	11.3559	bits/s/Hz
Energy Efficiency	1.2904	bits/s/Hz/W

The present implementation uses the corresponding power model defined in the simulation. Therefore, the same power assumptions are maintained across all beam-selection methods to ensure a fair performance comparison.

D. Greedy Baseline Results

The preliminary Greedy-only simulation produces the following baseline results.

For the considered configuration of 20 users, 16 antennas, 8 RF chains, and 3 hotspots, the Greedy beam-selection procedure achieves an average SINR of approximately 0.3089 dB and a spectral efficiency of 11.3559 bits/s/Hz, averaged over 15 channel seeds. The corresponding energy efficiency is 1.2904 bits/s/Hz/W under the adopted power-consumption model.

These values establish the classical Greedy baseline for the subsequent optimization stages. Since the Greedy algorithm maximizes the individual user-beam alignment gain rather than explicitly optimizing interference, the selected high-gain beams can still produce inter-user interference. Conse-

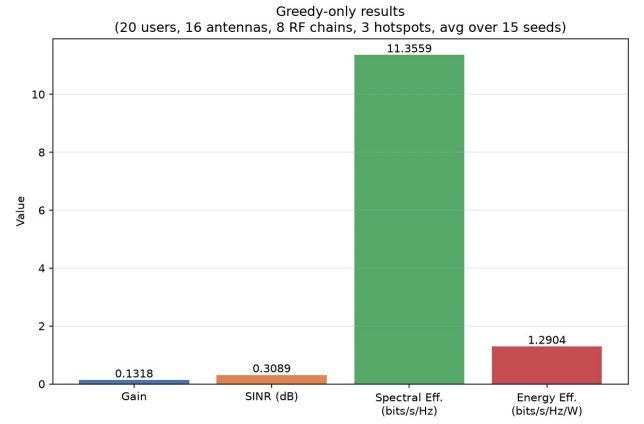


Fig. 2. Greedy-only baseline performance for the considered system configuration. The evaluated metrics are SINR, spectral efficiency, and energy efficiency. Results are averaged over 15 independent channel seeds.

quently, improving the beam-selection criterion provides an opportunity to improve SINR and spectral efficiency while maintaining or improving energy efficiency.

E. Performance Analysis

The three performance metrics provide complementary information about the quality of the selected beam assignments.

First, SINR measures the quality of the desired signal relative to inter-user interference and receiver noise. Since the Greedy algorithm selects beams primarily according to the gain matrix, its selection criterion does not directly account for interference between simultaneously selected users.

Second, spectral efficiency is directly related to SINR through (16). Therefore, improvements in SINR generally result in higher achievable spectral efficiency.

Third, energy efficiency considers both the achieved spectral efficiency and the total power consumption through (18). This metric is particularly relevant for hybrid beamforming because reducing the number of RF chains can reduce hardware power consumption while retaining high spectral efficiency.

For the present configuration, the Greedy algorithm therefore serves as a classical reference against which the subsequent beam-selection techniques can be evaluated.

F. Planned Performance Comparison

The Greedy implementation establishes the classical baseline for the next stages of the project. The same channel realizations and system parameters will subsequently be used to implement OMP and QAOA-based beam-selection methods.

The final comparison will focus on

$$\boxed{\text{SINR, Spectral Efficiency, Energy Efficiency}}. \quad (20)$$

The planned comparison is

$$\text{Greedy vs. OMP vs. QAOA.} \quad (21)$$

All methods will be evaluated using identical channel conditions, numbers of antennas, users, RF chains, candidate beams, and power assumptions to ensure a fair comparison.

V. CONCLUSION

This work developed a multi-user massive MIMO framework for beam-to-RF-chain assignment with the objective of improving SINR, spectral efficiency, and energy efficiency. A clustered geometric channel model was implemented to generate the channel characteristics of users distributed across multiple spatial hotspots, and a DFT-based beam codebook was constructed for a 16-antenna ULA. Based on the channel and beam responses, a user-beam gain matrix was constructed and a Greedy beam-selection algorithm was implemented to assign the available 8 RF chains to feasible user-beam pairs under one-user and one-beam constraints.

Preliminary simulation results for 20 users, 16 antennas, 8 RF chains, and 3 hotspots, averaged over 15 channel seeds, establish a baseline average SINR of 0.3089 dB, spectral efficiency of 11.3559 bits/s/Hz, and energy efficiency of 1.2904 bits/s/Hz/W under the adopted power model. The developed channel-model and beam-assignment framework provides the foundation for the next stage of the work, where OMP- and QUBO/QAOA-based beam-selection methods will be investigated and compared against the Greedy baseline using SINR, spectral efficiency, and energy efficiency.

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